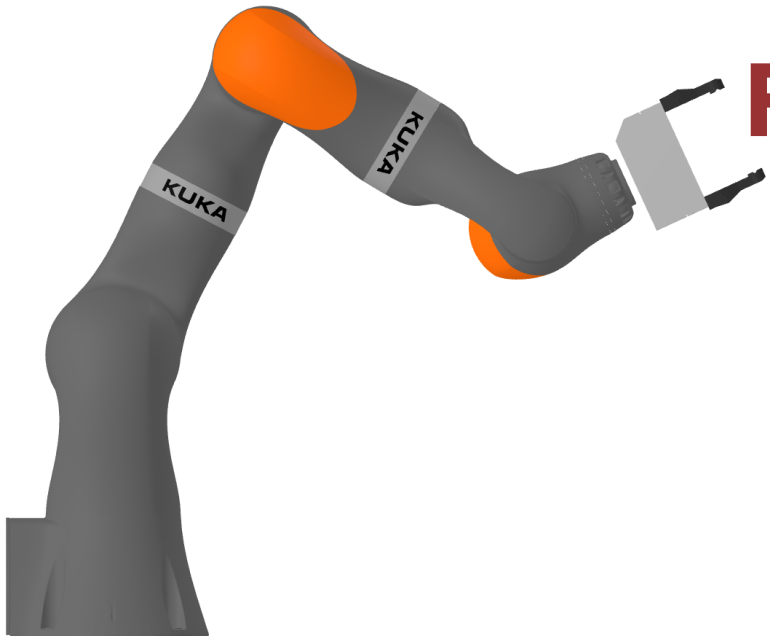


Force Control

MIT 6.4210/2

Robotic Manipulation

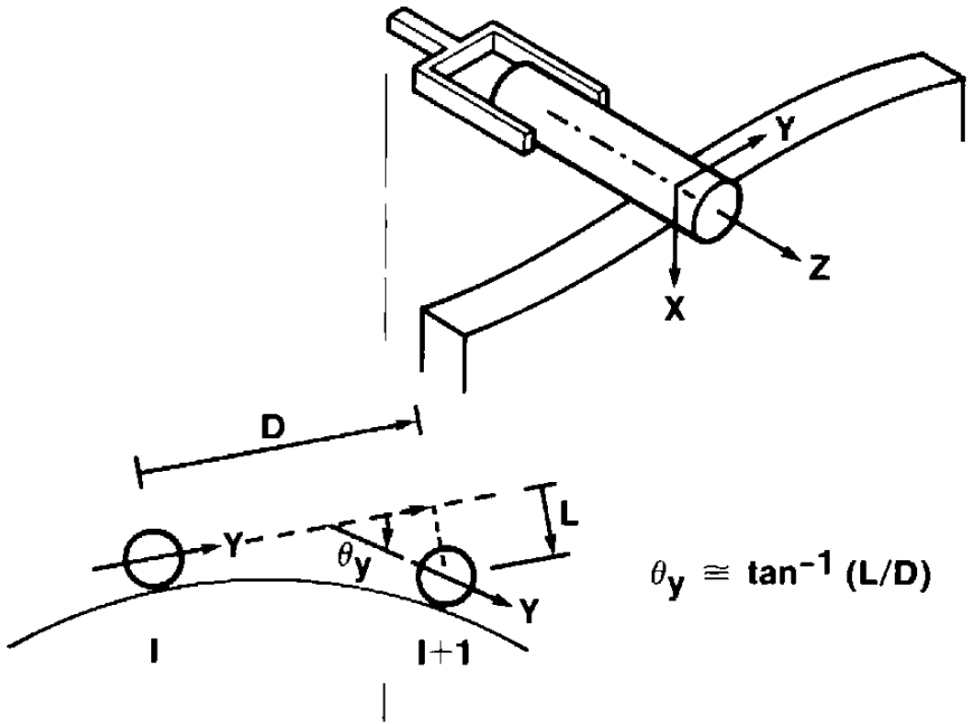
Fall 2025, Lecture 13

Massachusetts Institute of Technology


CHEEZ-IT
Original
JELL-O
Fruits
40% less

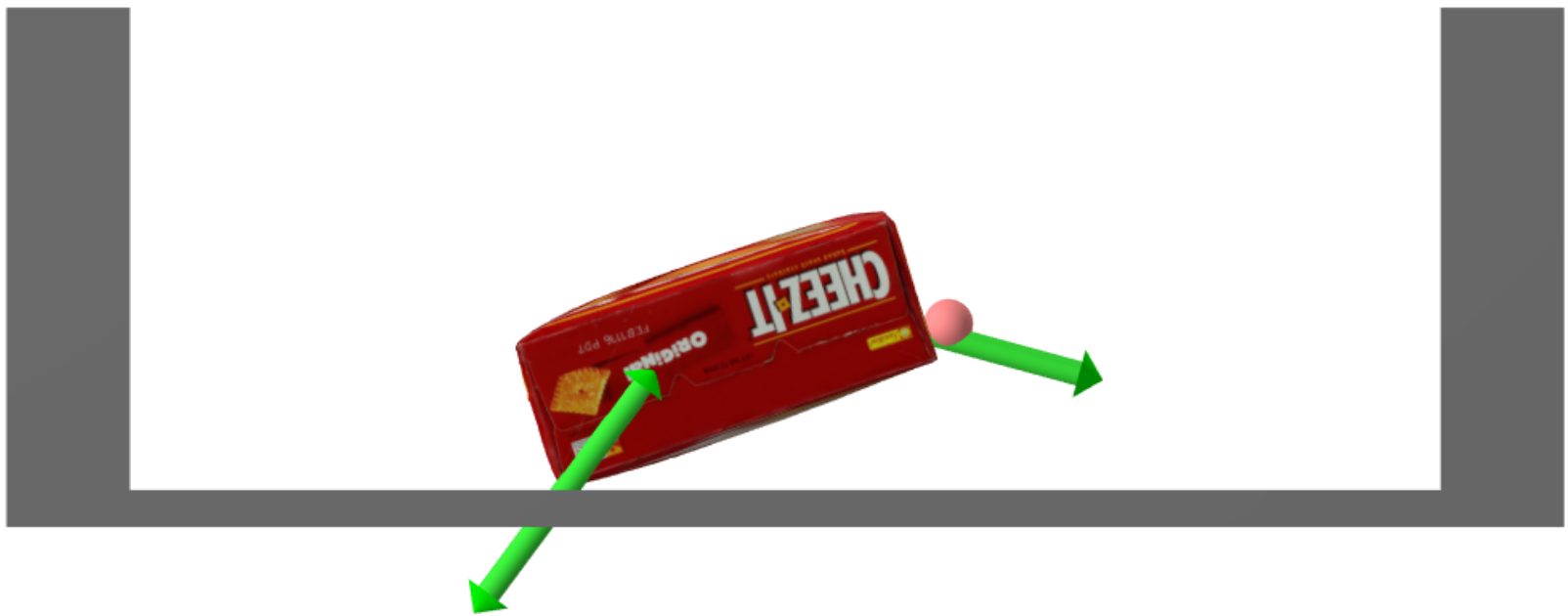
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EDGE FOLLOWING



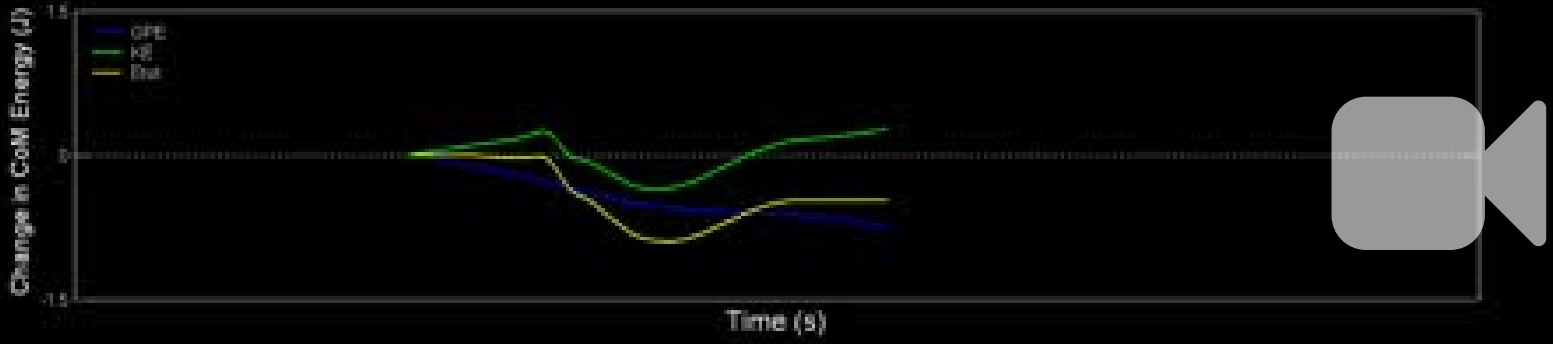
D. E. Whitney, “Historical Perspective and State of the Art in Robot Force Control,” *The International Journal of Robotics Research*, vol. 6, no. 1, pp. 3–14, Mar. 1987.

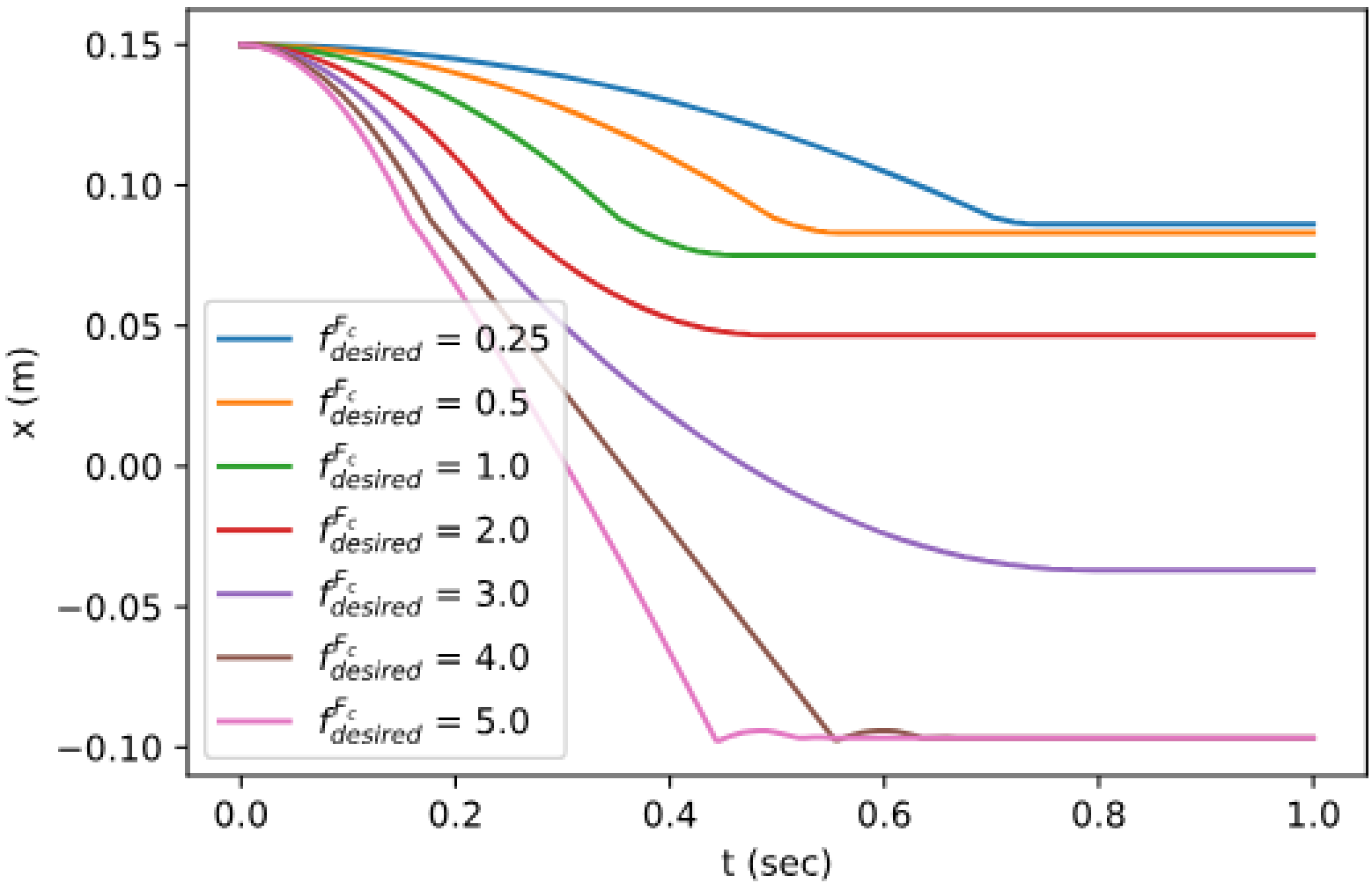
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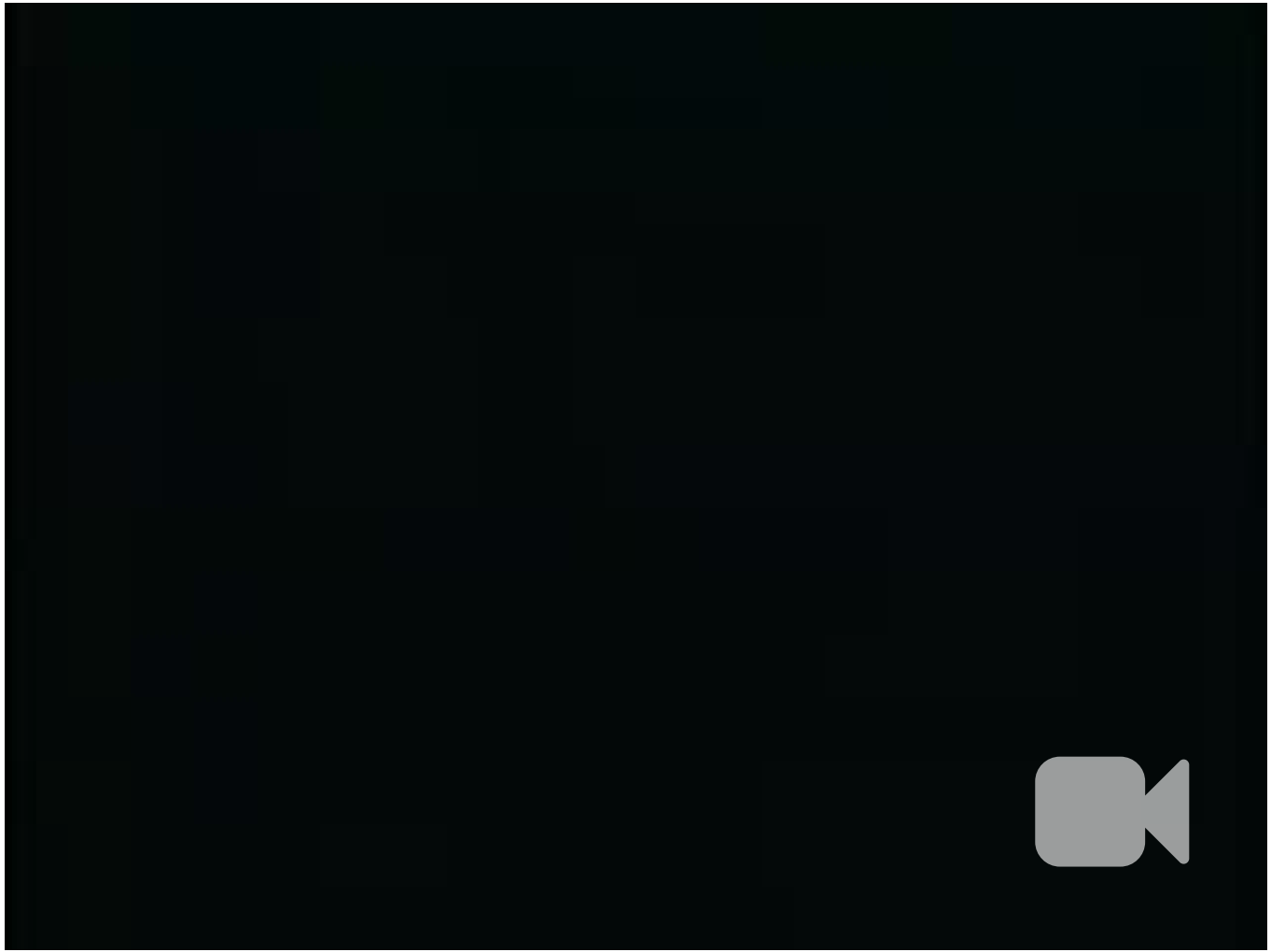

CHEEZ-IT



Monica A. Daley




0.15
0.10
0.05
0.00
−0.05
−0.10
x (m)
0.0
0.2
0.4
0.6
0.8
1.0
t (sec)
$f^{F_c}_{desired} = 0.25$
$f^{F_c}_{desired} = 0.5$
$f^{F_c}_{desired} = 1.0$
$f^{F_c}_{desired} = 2.0$
$f^{F_c}_{desired} = 3.0$
$f^{F_c}_{desired} = 4.0$
$f^{F_c}_{desired} = 5.0$



Inoue 1974 (at MIT AI Lab)

https://manipulation.mit.edu/clutter.html#spatial_force_algebra

https://www.youtube.com/embed/1Or0kym2uic?mute=1&start=5&enablejsapi=1

https://www.youtube.com/embed/dnUwqngH0bM?
start=35&mute=1&enablejsapi=1

https://www.youtube.com/embed/P-d-yTcxdzw?enablejsapi=1

USING COMPLIANCE IN LIEU OF SENSORY FEEDBACK FOR AUTOMATIC ASSEMBLY

by

SAMUEL HUNT DRAKE

SUBMITTED IN PARTIAL FULFILLMENT
OF THE REQUIREMENTS FOR THE
DEGREE OF

DOCTOR OF SCIENCE

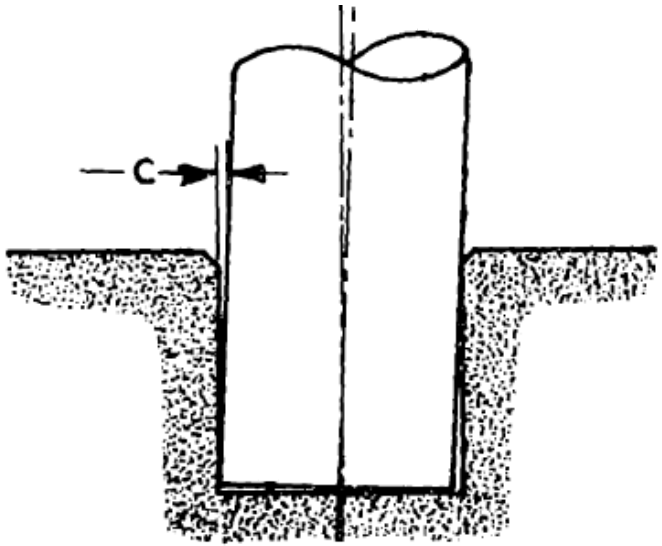
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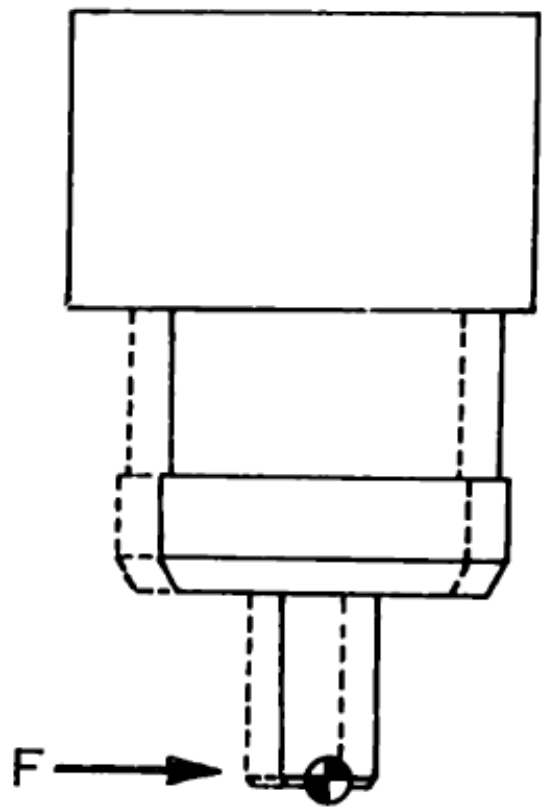
MASSACHUSETTS INSTITUTE OF TECHNOLOGY

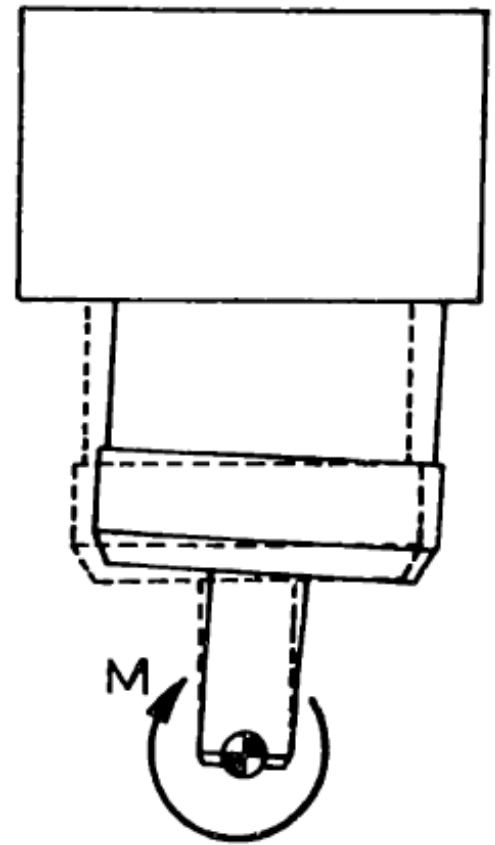
September, 1977

(c. February, 1978)

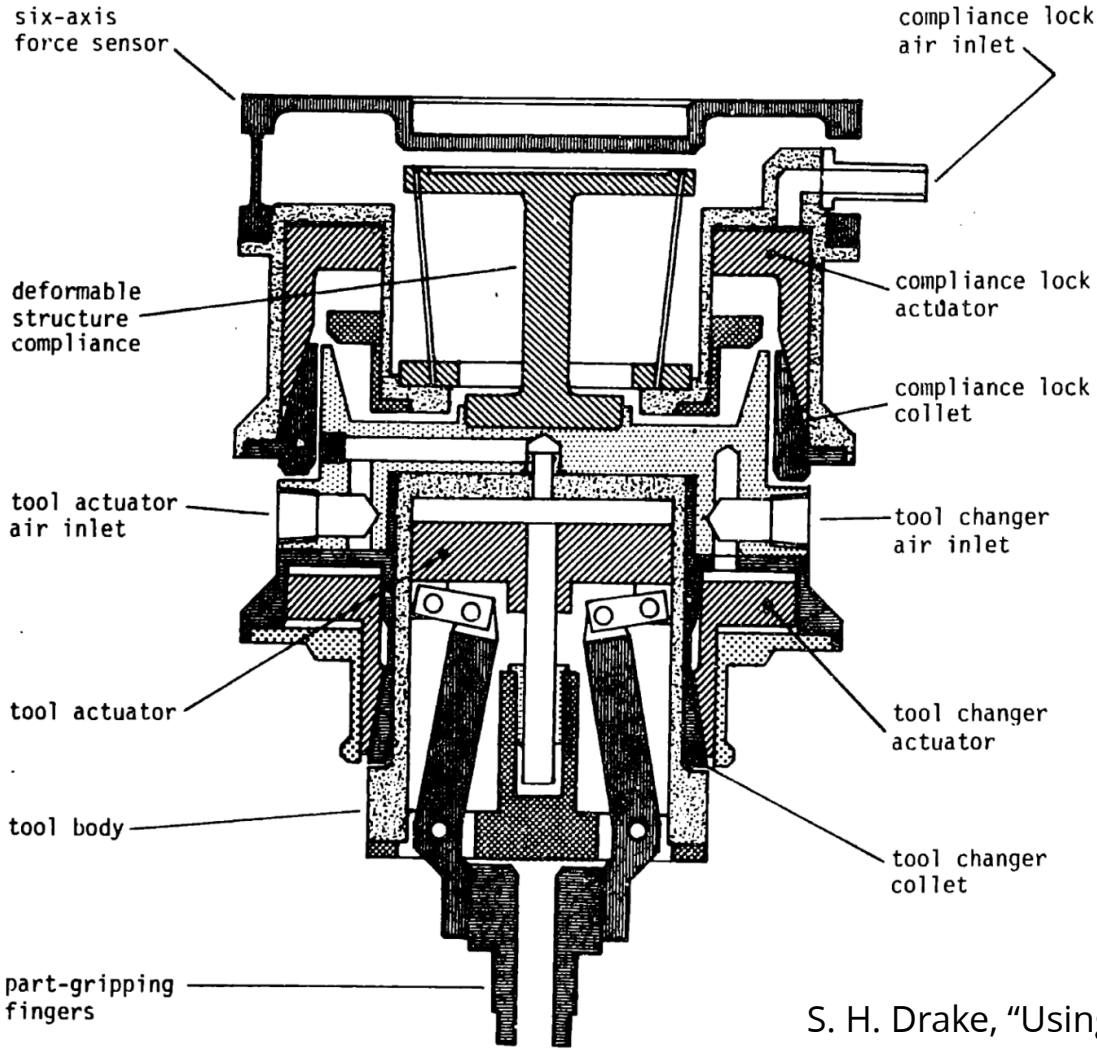
SAMUEL HUNT DRAKE

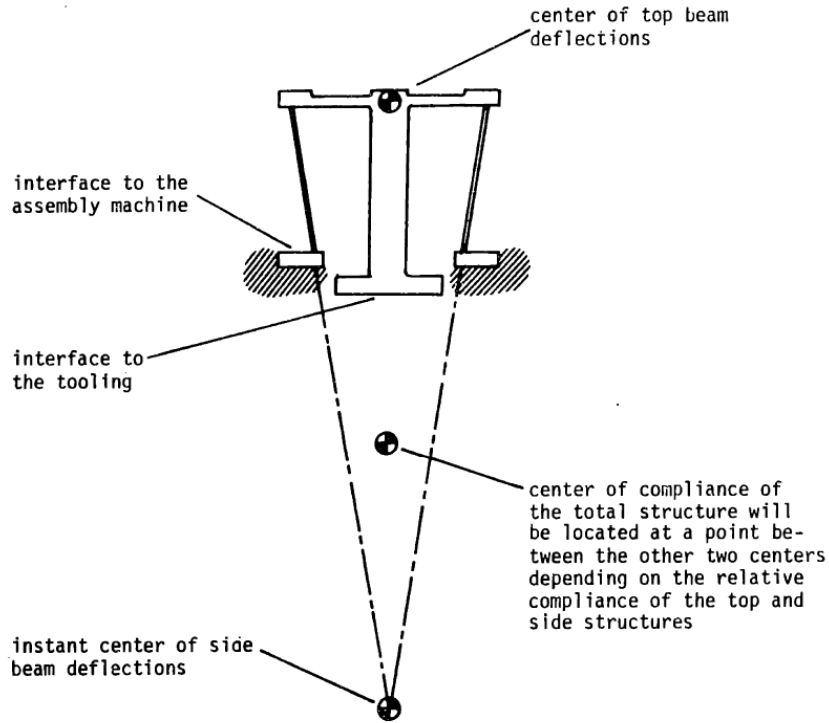






S. H. Drake, “Using compliance in lieu of sensory feedback for automatic assembly.,” Massachusetts Institute of Technology, 1978.[15]





S. H. Drake, “Using compliance in lieu of sensory feedback for automatic assembly.,” Massachusetts Institute of Technology, 1978.

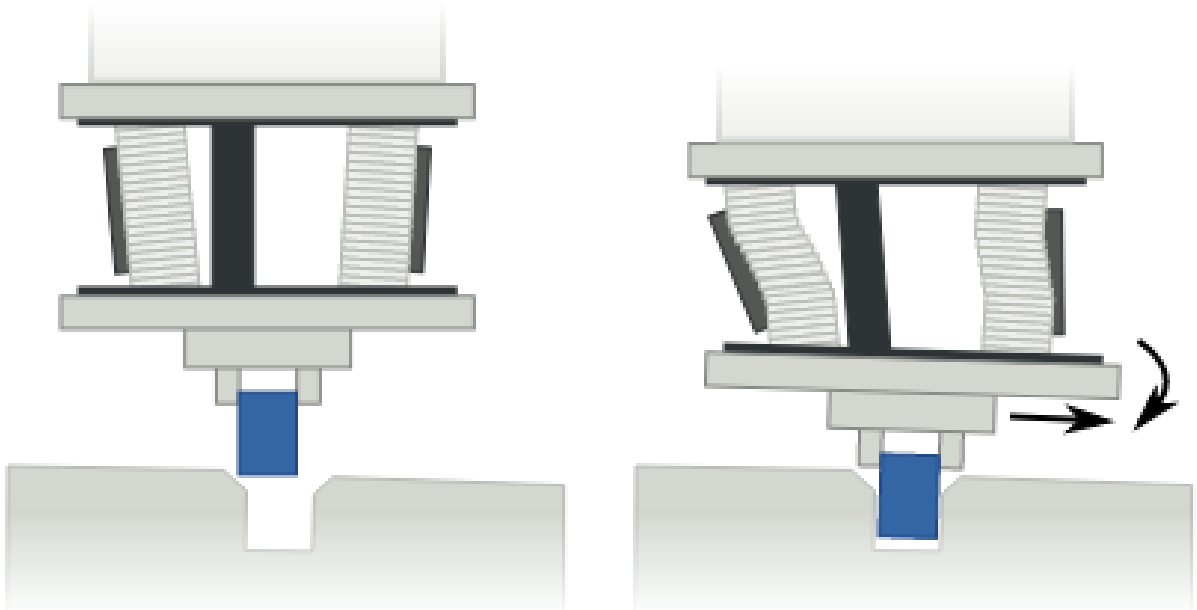


Image credit for Arne Nordmann, 2008

https://www.youtube.com/embed/qFTUFHGGZIg?mute=1&rel=0&enablejsapi=1